Pratik Kumar Mishra

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EDUCATION

Ph.D.

University of Toronto, Canada [Jan 2021 – Aug 2026]
Supervisors: Dr. Shehroz S. Khan, Dr. Alex Mihailidis
Thesis: Unsupervised Deep Learning Approaches for Detection of Behaviours of Risk in People with Dementia

MS in Computer Science and Engineering

Indian Institute of Technology Indore, India [Jul 2018 – Oct 2020]
Supervisor: Dr. Aruna Tiwari
Thesis: Minimum Variance Embedding-Based Least-Squares Methods for One-Class Classification

Bachelor of Technology in Computer Science and Engineering

Biju Patnaik University of Technology, India [May 2011 – May 2015]

RESEARCH

Journal Papers

1. Depth-Weighted Detection of Behaviours of Risk in People with Dementia using Cameras. **Pratik K. Mishra**, Irene Ballester, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, and Shehroz S. Khan. Biomedical Signal Processing & Control, 2026. [Paper]
2. Large Language Models Improve Scene Invariant Detection of Behaviours of Risk in Dementia Residential Care Across Multiple Surveillance Camera Views. **Pratik K. Mishra**, Babak Taati, Bing Ye, Kristine Newman, Alex Mihailidis, Andrea Iaboni, and Shehroz S. Khan. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2026. [Paper]
3. Examining Canadian Smart Home Technology Users' Perceptions of Privacy, Risks, Safety, and Data Collection Purposes: Cross-Sectional National Survey. Eva Victoria Di Nallo, Charlene H Chu, Zara Hasan, **Pratik K. Mishra**, Jacob Buchan, Tun Zheng, Andria Bianchi, Jesse Hoey, and Shehroz S Khan. JMIR Formative Research. (Accepted)
4. A Novel Multi-modal Sensor Dataset and Benchmark to Detect Agitation in People Living with Dementia in a Residential Care Setting. Shehroz S. Khan, **Pratik K. Mishra**, Bing Ye, Smit Patel, Kristine Newman, Alex Mihailidis, and Andrea Iaboni. ACM Transactions on Computing for Healthcare. 2024. [Paper]
5. Skeletal Video Anomaly Detection using Deep Learning: Survey, Challenges and Future Directions. **Pratik K. Mishra**, Alex Mihailidis, and Shehroz S. Khan. IEEE Transactions on Emerging Topics in Computational Intelligence. 2024. [Paper]
6. Privacy Protecting Behaviours of Risk Detection in People with Dementia using Videos. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, and Shehroz S. Khan. BioMed Eng OnLine. 2023. [Paper]
7. Unsupervised Deep Learning to Detect Agitation From Videos in People With Dementia. Shehroz S. Khan, **Pratik K. Mishra**, Nizwa Javed, Bing Ye, Kristine Newman, Alex Mihailidis, and Andrea Iaboni. IEEE Access. 2022. [Paper]
8. Minimum Variance-Embedded Auto-associative Kernel Extreme Learning Machine for One-class Classification. **Pratik K. Mishra**, C. Gautam, and A. Tiwari. Neural Computing and Applications. 2021. [Paper]
9. Graph-Embedded Multi-layer Kernel Ridge Regression for One-class Classification. C. Gautam, A. Tiwari, **Pratik K. Mishra**, S. Suresh, A. Iosifidis, and M. Tanveer. Cognitive Computation. 2021. [Paper]
10. Minimum Variance-Embedded Deep Kernel Regularized Least Squares Method for One-class Classification and Its Applications to Biomedical Data. C. Gautam, **Pratik K. Mishra**, A. Tiwari, B. Richhariya, H. M. Pandey, S. Wang, and M. Tanveer. Neural Network. 2020. [Paper]

Conference Proceedings

1. Scene Invariant Cross Camera Anomaly Detection of Behaviours of Risk in People with Dementia. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, and Shehroz S. Khan. International Joint Conferences on Artificial Intelligence Workshop, Montreal, Quebec. 2025. [Paper]
2. Empirical Thresholding on Spatio-temporal Autoencoders Trained on Surveillance Videos in a Dementia Care Unit. Shehroz S. Khan*, **Pratik K. Mishra***, Bing Ye, Kristine Newman, Andrea Iaboni, and Alex Mihailidis. 20th Conference on Robots and Vision, Montreal, Quebec. 2023. [Paper]
   * The first and second authors have equal contributions.

Posters and Abstracts

1. Detection of Agitation in People with Dementia using Different Video Camera Views. Pratik K. Mishra, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. Toronto Biomedical Engineering Conference, 2024.
2. Detection of Agitation in People with Dementia using Different Video Camera Views. Pratik K. Mishra, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. International Conference on Aging, Innovation and Rehabilitation, Toronto, Canada, 2024.
3. Privacy Protecting Detection of Agitation in People with Dementia using Videos. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. RehabWeek, Singapore, 2023.
4. Privacy Protecting Detection of Agitation in People with Dementia using Videos. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. International Conference on Aging, Innovation and Rehabilitation. 2023.
5. Detection of Behaviors of Risk exhibited by People with Dementia with Privacy Protecting Videos. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. Alzheimer's Association International Conference, Amsterdam, Netherlands. 2023. [Abstract]
6. Detection of Behaviors of Risk exhibited by People with Dementia with Privacy Protecting Videos. **Pratik K. Mishra**, Andrea Iaboni, Bing Ye, Kristine Newman, Alex Mihailidis, Shehroz S. Khan. AGE-WELL Annual Conference. 2022.
7. Automatic Detection of Behaviours of Risk in People with Dementia using Unsupervised Deep Learning. Pratik K. Mishra. Proceedings of the Canadian Conference on Artificial Intelligence. 2022. [Abstract]

TEACHING EXPERIENCES

Prospective Professors in Training

University of Toronto, Canada [Jan – Aug 2024]

Teaching Assistant

University of Toronto, Canada
Courses:

- BME205: Fundamentals of Biomedical Engineering [Jan - April 2024, 2025]
- BME1560: Artificial Intelligence for Biomedical Engineering [Sep - Dec 2022, 2024]

Teaching Assistant

Indian Institute of Technology Indore, India [Jul 2018 – Jun 2020]
Courses:

- CS 207: Database systems
- CS 304N: Computational Intelligence
- CS 202: Automata Theory and Logic
- CS 103: Computer Programming

WORK EXPERIENCES

Research Volunteer

KITE, Toronto Rehabilitation Institute , University Health Network, Canada [July 2020 – Dec 2020]

Associate System Engineer

IBM India Private Limited, India [Jul 2015 – Jul 2018]

ACADEMIC SERVICES

Program Committee Member
IJCAI-ECAI 2026 Special Track on AI and Health

Program Committee Member
Workshop on AI for Aging, Rehabilitation, and Intelligent Assisted Living, IJCAI 2025, IJCAI 2026

Organizing Committee Member
Workshop on AI for Aging, Rehabilitation, and Intelligent Assisted Living, ECML 2023, IJCAI 2024

Peer Reviewing
- ACM Computing Surveys
- IEEE Sensors
- IEEE Transactions on Image Processing
- Pattern Recognition
- Scientific Reports

AWARDS

Conference Travel Award
AGE-WELL | October 2024 | CAD 2000

Doctoral Completion Award
University of Toronto | September 2024 - August 2025 | CAD 6,000

EPIC-AT Fellowship
AGE-WELL and University of Toronto | September 2023 - August 2024 | CAD 17,000

SGS Conference Grant
University of Toronto | June 2023 | CAD 500

Conference Travel Award
AGE-WELL | October 2022 | CAD 1800

AGE-WELL-UofT FASE Graduate Student Award
AGE-WELL and University of Toronto | September 2022 - August 2023 | CAD 15,000

2021-22 Wildcat Graduate Scholarship
University of Toronto | September 2021 - August 2022 | CAD 10,000

GATE Scholarship
Government of India | July 2018 - June 2020 | INR 297,600

OTHER ACTIVITIES

- My research on privacy-protecting behaviours of risk detection in people with dementia using computer vision was featured by UHN Research.